

The Curious Case of SignMuon: To Error Feedback or not to Error Feedback?

Anonymous authors

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Abstract

The SignSGD gradient compression algorithm enables up to a $32\times$ reduction in communication volume, which is critical for federated learning in bandwidth-constrained environments. However, it often lags behind modern optimizers like Muon, which leverage the matrix structure of parameters to achieve superior convergence and performance. In this work, we propose SignMuon, an algorithm that applies sign compression to the Linear Minimization Oracle (LMO) update of the Muon optimizer. Our empirical results demonstrate that SignMuon achieves accuracy nearly on par with Muon while significantly outperforming SignSGD in both centralized and federated learning settings.

Keywords: federated learning, stochastic optimization, gradient compression, 1-bit quantization, matrix orthogonalization, Linear Minimization Oracle (LMO), SignSGD, Muon.

1 Introduction

Modern deep neural network (DNN) training methods require optimizers that not only provide high convergence rates but also minimize communication overhead in distributed systems. In federated learning, data are distributed across clients that communicate with a central server to train a global model. Each client possesses its own datasets and updates its parameters using a local optimizer. The goal of federated learning is to minimize the averaged loss function across all clients. Under limited network bandwidth, transmitting full model updates may significantly slow down the training process. Gradient compression methods are typically used to address this issue. Bernstein et al. (2018; 2019); Beznosikov et al. (2024)

For instance, SignSGD algorithm has demonstrated that even extreme compression (down to 1 bit per component) preserves the practical effectiveness of stochastic optimization and reduces the communication volume. This This data volume is reduced by a factor of approximately $32\times$ compared to standard SGD, with only minor degradation in model quality Bernstein et al. (2018; 2019). However, SignSGD does not account for the matrix structure of the parameters. In tasks with matrix weights, it treats matrices as one-dimensional vectors (flattening), ignoring their intrinsic two-dimensional tensor structure, which may negatively affect the training performance of DNN.

Recently, Muon has demonstrated that orthonormalized updates can stabilize DNN training and, in several cases, outperform standard adaptive optimization methods in terms of final solution accuracy Jordan et al. (2024); Bernstein and Newhouse (2024); Essential AI (2025). For matrix parameters, Muon implements a steepest descent step in the spectral norm. According to Jordan et al. (2024), the update direction is computed as the orthonormal matrix $\mathbf{UV}^\top$, where $\mathbf{U}$ and $\mathbf{V}$ are the singular-vector matrices obtained from the gradient decomposition. Based on the structure of the algorithm, Muon requires the transmission of full update matrices, which can be prohibitively computationally expensive for federated tasks.

In this work, we propose SignMuon¹, an optimization algorithm that applies sign compression to the Linear Minimization Oracle (LMO) update of Muon. SignSGD transmits the sign of the coordinates of the stochastic gradient direction. Our algorithm first computes an update direction using LMO, and then transmits only the sign of this direction. This approach allows for the matrix structure of the parameters to be accounted for and preserves the advantages of Muon, while reducing the amount

¹Code is available at: <https://github.com/intsystems/2026-Project-203/tree/main>

of transmitted information to one bit per coordinate. Consequently, SignMuon combines the fast convergence of structured optimizers with the communication efficiency of sign-based approaches.

The effectiveness of the proposed optimizer is evaluated on synthetic and practical tasks:

- the CIFAR-10 AirBench benchmark Jordan (2023);
- a synthetic least-squares optimization problem.

Our experiments investigate how effectively Muon-style structured update directions can be combined with the extreme communication compression characteristic of sign-based methods. Specifically, we analyze how transmitting only the update direction signal affects the accuracy and stability of training compared to using the full Muon update.

Experimental results show that applying sign compression to Muon directions reduces communication overhead (by $32\times$) while maintaining optimization rates. In the centralized setting, SignMuon consistently outperforms the baseline SignSGD method in both convergence speed and final accuracy. At the same time, the accuracy of the proposed 1-bit method is only $1 - 1.5\%$ lower than that of the Muon optimizer, while significantly outperforming basic signed algorithms. In the federated setting, integrating SignMuon with client momentum and Majority Vote aggregation method provides stable convergence and robustness to anomalous client updates even under severe communication constraints. Furthermore, we present an analytical example that formalizes the limitations of applying sign compression to LMO-based updates in the deterministic setting and for smooth convex optimization problems. The results of our experiments demonstrate that the proposed algorithm successfully combines LMO-based methods with sign compression without any significant loss in training accuracy.

2 Related Work

3 Problem Statement

Centralized Learning: We consider the stochastic optimization problem:

$$\min_{\mathbf{X} \in \mathcal{X}} \{f(\mathbf{X}) := \mathbb{E}_{\xi \sim \mathcal{D}}[f_{\xi}(\mathbf{X})]\}, \quad (1)$$

where $\mathcal{X}$ is the parameter space (e.g., $\mathbb{R}^d$ or $\mathbb{R}^{m \times n}$), functions $f_{\xi} : \mathcal{X} \rightarrow \mathbb{R}$ are potentially non-convex and non-smooth but continuously differentiable functions. Here, $f_{\xi}(\mathbf{X})$ is loss function of model parameterized by $\mathbf{X}$, computed at a data point ξ , sampled from the probability distribution $\mathcal{D}$.

Federated Learning: In modern machine learning, progress is increasingly driven by the ability to scale computational systems. State-of-the-art models rely on massive datasets and highly complex architectures, often requiring substantial computational resources and extended training times. As a result, the training process is formulated as an optimization problem in a distributed environment:

$$\min_{\mathbf{X} \in \mathcal{X}} \left\{ f(\mathbf{X}) := \frac{1}{N} \sum_{j=1}^N f_j(\mathbf{X}) \right\}, \quad f_j(\mathbf{X}) = \mathbb{E}_{\xi_j \sim \mathcal{D}_j}[f(\mathbf{X}, \xi_j)] \quad (2)$$

where $\mathcal{X}$ is the parameter space (e.g., $\mathbb{R}^d$ or $\mathbb{R}^{m \times n}$), $\mathbf{X} \in \mathcal{X}$ is model params, $N \geq 1$ is the number of clients, and $f_j(\mathbf{X})$ is the loss of the model $\mathbf{X}$ on the data $\mathcal{D}_j$, stored on client $j \in [N] := \{1, \dots, N\}$. In the federated learning, clients do local computations and periodically synchronizing with a global server. Since communication bandwidth is often limited, the volume of transmitted data becomes a major bottleneck in the training process Bernstein et al. (2019); Horváth et al. (2019). Several practical techniques for federated optimization and communication compression have also been studied in Beznosikov et al. (2024); Grunkowska et al. (2025); Ahn et al. (2025).

For theoretical optimization analysis in distributed environments, we rely on the assumption that the objective function f is L -smooth with respect to a general norm $\|\cdot\|$. Formally, for any $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^d$, the following condition holds:

$$\|\nabla f(\mathbf{X}) - \nabla f(\mathbf{Y})\|_* \leq L\|\mathbf{X} - \mathbf{Y}\|, \quad (3)$$

where $\|\cdot\|_*$ is the dual norm defined by $\|\mathbf{X}\|_* := \sup_{\|\mathbf{X}'\| \leq 1} \langle \mathbf{X}, \mathbf{X}' \rangle$ for all $\mathbf{X} \in \mathbb{R}^d$ Kovalev (2025). We note that, in practice, weaker smoothness assumptions such as the generalized (L_0, L_1) -smoothness condition may also be considered Pethick et al. (2026). Limited network bandwidth makes the transmission of full-precision 32-bit gradients prohibitively expensive, especially when training large-scale models. This motivates the study of communication-efficient optimization methods based on extreme gradient compression, where the communication budget is reduced to a single bit per transmitted coordinate Bernstein et al. (2018; 2019); Beznosikov et al. (2024).

Linear Minimization Oracle (LMO) is defined as an operator $A(\cdot)$ that provides the solution to a linear minimization problem over a feasible set:

$$A(\mathbf{X}) = \arg \min_{\mathbf{Y} \in \mathcal{S}} \langle \mathbf{X}, \mathbf{Y} \rangle. \quad (4)$$

where $\mathbf{X} \in \mathbb{R}^{m \times n}$ is the effective update direction, $\mathcal{S} \subset \mathbb{R}^{m \times n}$ is the convex set, and $\langle \mathbf{X}, \mathbf{Y} \rangle = \sum_{i,j} X_{ij} Y_{ij}$ is the standard matrix inner product Pethick et al. (2025). In particular, the constraints can be defined via the unit ball of a given norm:

$$A(\mathbf{X}) = \arg \min_{\mathbf{Y} \in \{\mathbf{Y} \in \mathcal{S} \mid \|\mathbf{Y}\| \leq 1\}} \langle \mathbf{X}, \mathbf{Y} \rangle. \quad (5)$$

Such an operator selects the direction of steepest descent with respect to the specified norm. The analytical and practical role of the LMO is discussed in detail in the literature on norm-constrained LMO optimizers Pethick et al. (2025); Kovalev (2025); Riabinin et al. (2025).

Muon optimizer uses the matrix structure of the gradient. Muon implements a steepest descent step in the spectral norm, where the update direction is obtained by orthogonalizing the gradient matrix Bernstein and Newhouse (2024). Let $\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ be the singular value decomposition (SVD) of the matrix $\mathbf{M} \in \mathbb{R}^{m \times n}$, where $\mathbf{U} \in \mathbb{R}^{m \times r}$ and $\mathbf{V} \in \mathbb{R}^{n \times r}$ are the orthonormal matrices of singular vectors, $\mathbf{\Sigma} \in \mathbb{R}^{r \times r}$ is the diagonal matrix of singular values, and $r = \text{rank}(\mathbf{M})$. Then, the LMO direction takes the following form:

$$A(\mathbf{M}) = \mathbf{U}\mathbf{V}^\top \in \mathbb{R}^{m \times n}. \quad (6)$$

That is, Muon selects an orthonormal update direction corresponding to the solution of the linear minimization problem induced by the spectral norm geometry. In practice, Muon utilizes a smoothed gradient estimate with momentum, and its update is defined as follows:

$$\begin{aligned} \mathbf{M}_t &= \mu \mathbf{M}_{t-1} + \mathbf{G}_t, \\ \mathbf{M}_t &= \mathbf{U}_t \mathbf{\Sigma}_t \mathbf{V}_t^\top, \\ \mathbf{D}_t &= \mathbf{U}_t \mathbf{V}_t^\top, \\ \mathbf{X}_{t+1} &= \mathbf{X}_t - \eta_t \mathbf{D}_t. \end{aligned} \quad (7)$$

In the original Muon implementation, the computation of $\mathbf{U}_t \mathbf{V}_t^\top$ is approximated using Newto-Schulz iterations and related polynomial schemes. It enables efficient matrix orthogonalization without an explicit SVD computation Amsel et al. (2026); Li and Hong (2025); Liu et al. (2025). Such orthogonalization produces more isotropic updates and prevents a small number of dominant singular directions from controlling the optimization process, which is particularly beneficial for training DNN Bernstein and Newhouse (2024); Essential AI (2025).

Despite its theoretical and practical advantages, Muon requires transmitting the full orthogonalized update matrix $\mathbf{D}_t$. This step makes Muon more expensive in federated learning settings. Our objective is to develop a communication-efficient optimization algorithm, that preserves the geometric properties of Muon’s LMO-based updates while reducing the communication cost to the level of SignSGD (1 bit per parameter). At the same time, the proposed method should retain the standard convergence guarantees for smooth non-convex optimization:

$$\min_{0 \leq t \leq T} \mathbb{E} [\|\nabla f(\mathbf{X}_t)\|_*^2] = \mathcal{O}(T^{-1/2}). \quad (8)$$

4 Theory

Consider the SignA family of algorithms, where A denotes any optimizer based on LMO directions. All methods in this family follow the same three-stage structure: momentum accumulation, computation of an LMO direction, and sign-compression of the resulting update.

Formally, a SignA method initializes the momentum buffer as $\mathbf{M}_0 = \mathbf{0}$, and at each iteration $t \geq 1$ performs the following sequence of updates:

$$\begin{aligned}
\mathbf{M}_t &= \mu \mathbf{M}_{t-1} + \mathbf{G}_t && \text{(Momentum accumulation)} \\
\tilde{\mathbf{M}}_t &= \begin{cases} \mathbf{G}_t + \mu \mathbf{M}_t & \text{(Nesterov momentum),} \\ \mathbf{M}_t & \text{(otherwise)} \end{cases} && \text{(Effective direction),} \\
\mathbf{D}_t &= A(\tilde{\mathbf{M}}_t), && \text{(LMO projection)} \\
\mathbf{s}_t &= \text{sign}(\mathbf{D}_t) \in \{\pm 1\}^{m \times n}, && \text{(Sign compression)} \\
\mathbf{X}_{t+1} &= \mathbf{X}_t - \eta_t \lambda_{\text{mult}} \mathbf{s}_t && \text{(Parameter update),}
\end{aligned} \tag{9}$$

where $\mathbf{X}_t \in \mathbb{R}^{m \times n}$ is the current model parameter matrix, $\mathbf{G}_t$ is its stochastic gradient, $\mu \in [0, 1]$ is the momentum coefficient, η_t is the learning rate, and $\lambda_{\text{mult}} > 0$ is an optional scaling factor (with $\lambda_{\text{mult}} = 1$ by default). The operator $A(\cdot)$ denotes a Linear Minimization Oracle (LMO) in a specified non-Euclidean norm, and the function $\text{sign}(\cdot)$ is applied element-wise to the resulting structured direction $\mathbf{D}_t$.

The proposed SignMuon algorithm is a particular instance of the SignA family. It combines the orthogonalized LMO directions of the Muon optimizer with the extreme sign-based compression strategy of SignSGD. In the following sections, we describe two main variants of SignMuon: a centralized version and a federated version.

4.1 Centralized Learning

In the centralized setting, SignMuon is applied to the matrix-valued parameters of a neural network. The practical implementation takes into account the tensor structure of model parameters. The optimization process is divided into two separate streams: matrix parameters (weights of convolutional and fully connected layers, $n_{\text{dim}} \geq 2$) are updated using the SignMuon LMO-based procedure, whereas vector parameters (bias, Batch Normalization layer, $n_{\text{dim}} = 1$) as well as the final classification layer are optimized using the standard AdamW optimizer. This design was originally introduced in the empirical description of Muon by Jordan et al. (2024). The first theoretical convergence result was derived by Li and Hong, who analyzed the smooth non-convex setting Li and Hong (2025) and subsequently formalized within the Gluon framework Riabinin et al. (2025).

For matrix parameters, the key component of SignMuon is the computation of an orthonormal LMO direction $\mathbf{D}_t \approx \mathbf{U}_t \mathbf{V}_t^\top$. To avoid the computational cost of performing a full singular value decomposition (SVD) at every optimization step, the orthogonalization is approximated using a 5-order Newton-Schulz iteration. In an implementation, the Newton-Schulz coefficients are chosen as $a = 3.4445$, $b = 4.7750$, $c = 2.0315$. The main difference between SignMuon and the classical SignSGD algorithm lies in the object being compressed. In SignSGD, sign compression is applied directly to the stochastic gradient. In contrast, SignMuon first constructs a structured update direction $\mathbf{D}_t$, corresponding to the Muon LMO step and only then applies the elementwise sign operator to obtain $\text{sign}(\mathbf{D}_t)$. Consequently, SignMuon preserves the geometry of orthonormal updates while reducing the transmitted information to one bit per component. The complete update procedure for matrix parameters is presented in Appendix A.1.

4.2 Federated Learning

In federated learning, the training data are distributed across N clients. Each client j possesses a local dataset $\mathcal{D}_j$ and computes a local loss function $f_j(\mathbf{X})$. The goal of the training process is to minimize the averaged loss function across all clients (2). Since clients do not exchange raw data and communicate only through messages sent to the server, communication efficiency becomes a critical concern in federated optimization.

In the federated version of SignMuon, the server acts as an aggregation node, while the main optimization procedure is performed locally on the clients. At the beginning of communication round t , each client j receives a copy of the current global model. Rather than performing local parameter updates, the client computes an averaged stochastic gradient by accumulating gradients over n_{steps} local mini-batches. This gradient accumulation procedure reduces the variance of the stochastic gradient estimate before the application of the LMO operator. Formally, the accumulated gradient is

defined as:

$$\mathbf{G}_t^{(j)} \leftarrow \frac{1}{n_{\text{steps}}} \sum_{i=1}^{n_{\text{steps}}} \nabla f(\mathbf{X}_t; \xi_{t,i}^{(j)}).$$

In accordance with the general SignA framework (9), each client maintains its own local momentum buffer $\mathbf{M}_t^{(j)}$. After computing the accumulated gradient, the client updates the momentum estimate, applies the LMO (implemented via Newton-Schulz orthogonalization), and then performs element-wise sign compression:

$$\begin{aligned} \mathbf{M}_t^{(j)} &= \mu \mathbf{M}_{t-1}^{(j)} + \mathbf{G}_t^{(j)} \\ \mathbf{s}_t^{(j)} &= \text{sign}(A(\mathbf{M}_t^{(j)})) \in \{\pm 1\}^{m \times n} \end{aligned}$$

where $t \in [0, \text{rounds} - 1]$, $A(\cdot)$ denotes the Muon operator implemented via Newton-Schulz orthogonalization with coefficients $a = 3.4445$, $b = 4.7750$, $c = 2.0315$). Thus, each client transmits only 1 bit per matrix parameter component to the server.

On the server, messages received from the client are aggregated using a majority vote. The aggregated sign is denoted by $\mathbf{s}_t^{\text{agg}}$, then it is given by the formula:

$$\mathbf{s}_t^{\text{agg}} = \text{sign} \left(\sum_{j=1}^N \mathbf{s}_t^{(j)} \right). \quad (10)$$

The majority-vote operation ensures that the resulting sign $\mathbf{s}_t^{\text{agg}}$ is equal to $+1$ or -1 in each component and corresponds to the “majority vote” of the clients. Since momentum had already been taken into account at the clients’ level, the server directly updates the global model of matrix parameters:

$$\mathbf{X}_{t+1} = \mathbf{X}_t - \eta_t \mathbf{s}_t^{\text{agg}}.$$

A special treatment is applied to the final classification layers of the model. These parameters are not updated using the SignMuon rule, instead they are optimized using AdamW. The complete federated SignMuon procedure is presented in Appendix A.2.

4.3 Federated Learning with Error Feedback

Although Federated SignMuon demonstrates strong empirical stability, sign-based compression inevitably introduces bias into the descent direction. To compensate for quantization error and ensure strong convergence guarantees, we incorporate an Error Feedback method into our approach, adapted for LMO optimizers as described in a recent paper by Gruntkowska et al. Gruntkowska et al. (2025).

In this Federated EF-SignMuon variant, the orthogonalization procedure (LMO) is moved from the clients to the server. Instead of computing the LMO direction locally, each client evaluates the difference between its current local momentum estimate and the previous gradient estimate, and then applies sign compression to this residual. The server aggregates the compressed updates, reconstructs a global gradient estimate, and only then applies the Muon operator. The complete algorithm is presented in Appendix A.3.

The transmission of an additional scalar $\alpha_t^{(j)}$ for each layer introduces insignificant communication overhead, keeping the overall communication cost at ≈ 1 bit per parameter. Thus, both proposed federated SignMuon architectures: the baseline majority-vote scheme and the error-feedback variant – effectively address the communication bottleneck in federated learning. They preserve the geometric benefits of Muon-style matrix orthogonalization while maintaining an extremely low communication budget.

4.4 Theoretical limitations

The hypotheses presented in Appendix A.4 rely on the assumption that the direction of steepest descent in the spectral norm (the LMO step of the Muon algorithm) remains a valid descent direction even after applying extreme sign compression. However, this is not always the case.

The classic gradient descent algorithm guarantees that anti-gradient direction $(-\mathbf{G}_t)$ is a descent direction. In the SignMuon algorithm, the step direction is defined by the matrix $\mathbf{s}_t = \text{sign}(\mathbf{U}_t \mathbf{V}_t^\top)$.

For the algorithm to guarantee convergence, the following descent condition must be met:

$$\langle \mathbf{G}_t, \text{sign}(\mathbf{U}_t \mathbf{V}_t^\top) \rangle > 0. \quad (11)$$

This raises the question: does spectral orthogonalization guarantee that condition (11) holds for arbitrary matrices? We formalize this question in the following theorem.

Theorem 1. *There exist a function $f(\mathbf{W})$ and a matrix dimension $N \geq 4$ such that the SignMuon algorithm diverges.*

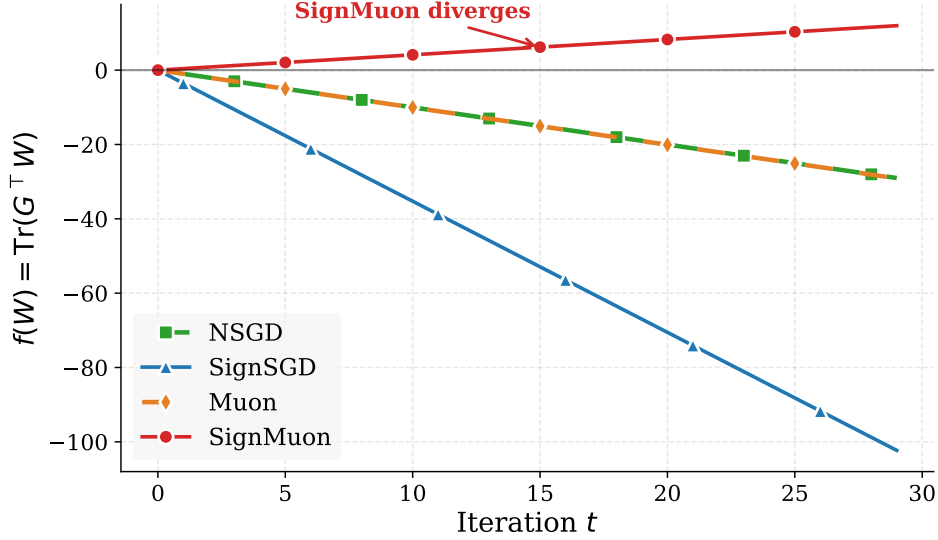


Рис. 1: Comparison of optimization trajectories for the linear objective function $f(\mathbf{W}) = \text{Tr}(\mathbf{G}^\top \mathbf{W})$, demonstrating the strict divergence of SignMuon, while other methods successfully minimize the function.

A proof by counterexample is provided in Appendix A.5. The existence of such an example demonstrates that SignMuon does not provide any global convergence guarantees for functions. Therefore, to explain its successful empirical performance on practical deep learning problems, where the algorithm achieves high accuracy, it is necessary to introduce additional assumptions.

5 Experiments

The goal of these experiments is empirical validation of algorithm SignMuon as an optimizer that combines the fast convergence properties of LMO-based methods with the communication efficiency of sign compression. The study consists of several parts, including evaluation in both centralized and federated settings, as well as validation on a synthetic least-squares optimization problem.

5.1 Smooth Convex Optimization Problem (Synthetic Experiment)

To evaluate the effect of matrix structure on the optimization process (without the influence of stochastic noise or the complexity of DNN architectures) we begin our experiments with a deterministic L -smooth convex optimization problem. We consider the minimization a quadratic form:

$$F(\mathbf{X}) = \frac{1}{2} \|\mathbf{M}^{1/2} \mathbf{X} \mathbf{N}^{1/2}\|_F^2 = \frac{1}{2} \langle \mathbf{X}, \mathbf{M} \mathbf{X} \mathbf{N} \rangle \rightarrow \min_{\mathbf{X} \in \mathbb{R}^{m \times n}}, \quad (12)$$

where the parameter matrix $\mathbf{X} \in \mathbb{R}^{m \times n}$ has dimension $m = n = 500$, which is representative of the size of parameter matrices in deep neural networks. The matrixes $\mathbf{M} \in \mathbb{S}_+^m$ and $\mathbf{N} \in \mathbb{S}_+^n$ are symmetric positive semidefinite. To generate the problem instance, we randomly construct the spectra of $\mathbf{M}$ and $\mathbf{N}$ by sampling their eigenvalues uniformly from the interval $(0, 1)$, which guarantees

a smoothness constant $L \leq 1$ with respect to the Frobenius norm. The parameter matrix $\mathbf{X}_0$ is initialized with elements which sampled independently from a normal distribution $\mathcal{N}(0, 0.01)$.

Поскольку теоретические оценки скорости сходимости зависят от выбора нормы и констант гладкости, будем использовать эмпирический критерий эффективности: алгоритмы оцениваются по минимальному количеству итераций, необходимых для достижения порогового значения функции потерь $F(\mathbf{X}) \leq 0.001$, максимальное число итераций $T_{\max} = 5000$. Для поиска оптимальных гиперпараметров оптимизаторов был проведен сеточный поиск (Grid Search) по learning rate η и коэффициенту импульса μ для каждого из пяти рассматриваемых оптимизаторов:

Таблица 1: Hyperparameter search grid for the synthetic convex problem.

Algorithm	Learning Rate (η)	Momentum (μ)
SignMuon	$[1 \cdot 10^{-4}, 1 \cdot 10^{-3}]$ with step $1 \cdot 10^{-4}$	$\{0.1, 0.2, \dots, 0.9, 0.95\}$
Muon	$[5 \cdot 10^{-3}, 2 \cdot 10^{-2}]$ with step $1 \cdot 10^{-3}$	$\{0.1, 0.2, \dots, 0.9, 0.95\}$
SignSGD	$[5 \cdot 10^{-5}, 2 \cdot 10^{-4}]$ with step $1 \cdot 10^{-5}$	$\{0.1, 0.2, \dots, 0.9, 0.95\}$
SGD	$[1 \cdot 10^{-2}, 1 \cdot 10^{-1}]$ with step $1 \cdot 10^{-2}$	$\{0.1, 0.2, \dots, 0.9, 0.95\}$
Adam	$[1 \cdot 10^{-2}, 1 \cdot 10^{-1}]$ with step $1 \cdot 10^{-2}$	—

Результаты сеточного поиска и итоговое количество итераций до достижения необходимой точности в 0.001 представлены в Таблице 2. Из результатов видно, что алгоритм Muon достиг оптимума за 1122 итерации, что сопоставимо с классическими методами первого порядка (Adam, SGD). Стоит отметить, что алгоритм SignMuon достигает целевого значения функции потерь за 2104 итерации, значительно превосходя базовый алгоритм SignSGD (3120 итераций). Динамика целевой функции и нормы градиента в процессе оптимизации проиллюстрированы в Приложении на Рисунке 2.

Таблица 2: Сравнение оптимизаторов на гладкой выпуклой задаче (уравнение 12). Показаны оптимальные гиперпараметры, обеспечивающие достижение $F(\mathbf{X}) \leq 0.001$ за наименьшее число итераций.

Algorithm	Learning rate (η)	Momentum (μ)	Iterations to 0.001
SignMuon	0.0002	0.2	2104
Muon	0.0065	0.1	1122
SignSGD	0.00015	0.8	3120
SGD	0.1	0.95	972
Adam	0.07	-	202

5.2 Централизованное обучение

В данном эксперименте мы сравниваем динамики сходимости и итоговые точности (Accuracy) алгоритма SignMuon с классическими оптимизаторами Muon, SignSGD, SGD и Adam при фиксированном количестве эпох. Одной из задач эксперимента является демонстрация того, что SignMuon обеспечивает высокое качество классификации, сопоставимое с Muon, при этом превосходит по точности классический алгоритм SignSGD при идентичной коммуникации (1 бит на параметр).

Эксперименты проводились на датасете CIFAR-10: стандартный датасет для задачи классификации изображений. Содержит 60 000 цветных изображений 10 классов. Каждое изображение представлено в виде тензора размерности $3 \times 32 \times 32$, где первый индекс соответствует количеству цветовых каналов (RGB). Для централизованного обучения датасет делится на обучающую и тестовую выборки. В федеративном сценарии обучающая выборка делится между узлами.

В качестве базовой модели была использована архитектура ResNet-18, адаптированная под изображения малого разрешения за счет использования начальной свертки 3×3 без макс-пулинга. Сеть включает четыре блока остаточных слоев (Residual Blocks) с пакетной нормализацией

(BatchNorm), завершаясь слоем регуляризации Dropout и линейным классификатором. Как было сказано ранее обучение происходило при фиксированном количестве эпох $T_{max} = 75$. В процессе обучения learning rate варьируется, используя косинусный планировщик скорости обучения (CosineAnnealingLR). Обновление learning rate η_t на эпохе t осуществляется по следующей формуле:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos \left(\frac{t}{T_{\max}} \pi \right) \right), \quad (13)$$

где $\eta_{\max}$ – начальный learning rate алгоритма, $\eta_{\min} = 0$ – минимальное значение (в нашей конфигурации), а $T_{\max}$ соответствует общему числу эпох обучения.

Результаты эксперимента и значения подобранных гиперпараметров указаны в Таблице 3. Динамика целевой функции и точности алгоритмов в процессе обучения проиллюстрированы в Приложении А.7.

Таблица 3: CIFAR10 - Централизованное обучение

Dataset	Optimizer	Epoch	Batch	LR	LR Aux	Mom	Train Acc	Test Acc
CIFAR-10	SignMuon	75	128	0.001	0.0001	0.9	99.69%	92.55%
	Muon			0.015	0.001		99.94%	93.70%
	SGD						99.98%	93.93%
	SignSGD			0.001	0.0001		93.19%	89.11%
	Adam						99.77%	93.03%

5.3 Федеративное обучение

Следующим этапом исследования стала оценка эффективности SignMuon в сценарии федеративного обучения (Federated Learning), где передача полных 32-битных матриц градиентов является узким местом. Цель экспериментов данного этапа – подтвердить, что экстремальное сжатие ортогонализированных обновлений позволяет достичь точности, сопоставимой с алгоритмом Muon, при экстремальном снижении объема передаваемой информации.

Для проведения тестов была реализована федеративная архитектура с центральным сервером агрегации. Чтобы обеспечить сопоставимость результатов с предыдущими этапами работы, использовалась та же сверточная нейронная сеть (CNN2), включающая два сверточных блока с BatchNorm и MLP-классификатор. Набор данных CIFAR-10 подвергался однородному (*homogeneous*) разбиению между узлами. В рамках данного сценария мы исследовали как базовую схему Federated SignMuon (algorithm 2), так и модификацию с механизмом учета ошибки Federated EF-SignMuon (algorithm 3). Каждый раунд обучения включал этап локального накопления градиента на узлах с последующей передачей данных, прошедших знаковую компрессию, на сервер для обновления глобальной модели.

Для системного анализа свойств предложенных алгоритмов данный этап экспериментального исследования был разделен на несколько подэкспериментов. В первую очередь необходимо было оценить, как масштаб федеративной сети влияет на стабильность знаковой оптимизации и итоговую точность модели. С этой целью было проведено детальное изучение базового алгоритма SignMuon при различном количестве активных узлов в системе

Эксперимент 1: Масштабируемость SignMuon по числу узлов. Исследовалась зависимость числа участвующих узлов на качество обучения с алгоритмом SignMuon. Число узлов $N \in [1, 3, 5, 7, 9]$ при 2000 раундах обучения и одном локальном шаге на узле. Результаты показали рост точности классификации с увеличением числа узлов. Данный эффект объясняется улучшением качества агрегации через majority vote при большем числе независимых оценок знака градиента. Результаты данного эксперимента приведены в Приложении А.8.

Подтвердив положительное влияние количества узлов на стабильность алгоритма, мы перешли к прямому сравнению SignMuon с существующими алгоритмами. Для этого была выбрана конфигурация из трех узлов, позволяющая оценить эффективность алгоритмов в условиях ограниченных ресурсов федерации.

Эксперимент 2: Сравнение разных алгоритмов при числе узлов $N=3$. Проводилось сравнение шести оптимизаторов в сценарии с 3 узлами, 2000 раундами и одним локальным шагом. Результаты данного эксперимента приведены в Приложении А.9.

Однако потенциал знаковых алгоритмов оптимизации наиболее полно раскрывается при увеличении масштаба сети и объема локальных вычислений. Поэтому на заключительном этапе тестирования мы увеличили количество узлов до десяти.

Эксперимент 3: Сравнение разных алгоритмов при числе узлов $N=10$. Те же шесть оптимизаторов тестировались в условиях 10 узлов, 2000 раундов и трех локальных шагов на узле. В данном режиме SignMuon достиг качества Muon. Вывод эксперимента: при достаточном числе узлов ($N \geq 10$) алгоритм SignMuon обеспечивает качество, сопоставимое с Muon, при этом сокращая объем передаваемых данных в ~ 32 раза за счет передачи только знаков градиентов вместо полных значений. Результаты данного эксперимента приведены в Таблице 4, а также на Рисунке 7.

Таблица 4: CIFAR10 – Федеративное обучение. Эксперимент 3. Сравнение разных алгоритмов при числе узлов $N=10$.

Dataset	Optimizer	Rounds	Clients	Steps	Batch	LR	Mom	Test Acc
CIFAR-10	SignMuon	2000	10	3	64	0.001	0.9	84.57%
	Muon					0.05		85.22%
	SignSGD					0.001		78.98%
	SGD					0.03		81.38%
	Adam					0.001		78.36%
	SignMuon_EF					0.02		82.31%

6 Conclusion

В данной работе был предложен алгоритм SignMuon – метод оптимизации на основе LMO, который объединяет структурные преимущества матричной ортогонализации с экстремальной знаковой компрессией для коммуникационной эффективности. Наш подход объединил принципы работы современных оптимизаторов, таких как Muon, и семейство алгоритмов SignA, обеспечивая снижение объема передаваемых данных в 32 раза. Мы разработали аналитическую базу для исследования сходимости SignMuon на гладкой выпуклой задаче. В данном этапе исследования мы убедились, что LMO-обновление сохраняет свои геометрические преимущества даже после 1-битного сжатия (SignMuon значительно превосходит SignSGD в скорости сходимости до целевой точности по итерациям). Экспериментальные результаты на наборах данных MNIST и CIFAR-10, а также в сценариях федеративного обучения подтверждают практическую применимость предложенного метода: SignMuon демонстрирует точность, сопоставимую с алгоритмом Muon, и существенно превосходит классический SignSGD.

Несмотря на то, что данная работа предлагает новый комбинированный алгоритм, она также подсвечивает новые направления для будущих исследований. Наш анализ в рамках ?? выявил ограничение: в отличие от стандартного градиентного спуска, сочетание LMO и знаковой компрессии может приводить к расходимости в специфических условиях доминирующего сингулярного шума. Кроме того, наши текущие теоретические гарантии опираются на предположение о точном вычислении LMO-направления, в то время как на практике используются полиномиальные приближения с помощью метода Ньютона-Шульца.

Таким образом, мы добились существенных результатов в построении структурированного и коммуникационно-эффективного алгоритма оптимизации. Тем не менее, данная область остается открытой, оставляя множество путей для будущих исследований.

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A Appendix

A.1 Centralized algorithm

Algorithm 1 SignMuon

Input: Initial model $\mathbf{X}_0$, initial gradient $\mathbf{G}_0$, momentum coefficient μ , learning rate η_t , $\lambda_{\text{mult}} = 1$, number of Newton-Schulz iterations n_{steps}

Output: Updated model $\mathbf{X}$

```

1:  $\mathbf{M}_0 \leftarrow 0$ 
2: for  $t = 0$  to  $(T - 1)$  do
3:   Stochastic gradient  $\mathbf{G}_t$ 
4:    $\mathbf{M}_t \leftarrow \mu \mathbf{M}_{t-1} + \mathbf{G}_t$  ▷ Momentum accumulation
5:    $\tilde{\mathbf{M}}_t = \begin{cases} \mathbf{G}_t + \mu \mathbf{M}_t, & \text{(Nesterov momentum),} \\ \mathbf{M}_t, & \text{(otherwise)} \end{cases}$ 
6:    $\mathbf{Y} \leftarrow \text{ReshapeTo2D}(\tilde{\mathbf{M}}_t)$  ▷ Flatten tensor to matrix  $m \times n$ 
7:   for  $k = 1$  to  $n_{\text{steps}}$  do ▷ Newton-Schulz orthogonalization
8:      $\mathbf{A} \leftarrow \mathbf{Y} \mathbf{Y}^\top$ 
9:      $\mathbf{Y} \leftarrow a \mathbf{Y} - b \mathbf{A} \mathbf{Y} + c \mathbf{A}^2 \mathbf{Y}$ 
10:  end for
11:   $\mathbf{D}_t \leftarrow \text{ReshapeToOriginal}(\mathbf{Y})$  ▷ Reshape back to original tensor
12:   $\mathbf{s}_t \leftarrow \text{sign}(\mathbf{D}_t)$  ▷ Sign compression
13:   $\mathbf{X}_{t+1} \leftarrow \mathbf{X}_t - \eta_t \mathbf{s}_t$  ▷ Update parameters
14: end for

```

A.2 Federated Algorithm

Algorithm 2 Federated SignMuon

Input: Initial model $\mathbf{X}_0$, clients number N , rounds number rounds, local steps number n_steps, learning rate for the main layers η_t , learning rate for the last layer η_t^{aux} , momentum coefficient μ

Output: Updated global model $\mathbf{X}$

```

1: for  $j = 1$  to  $N$  do
2:    $\mathbf{M}_0^{(j)} \leftarrow 0$  ▷ Initialization of local buffers on clients
3: end for
4: Initialize AdamW for last_layer with learning rate  $\eta_0^{\text{aux}}$ 
5: for  $t = 0$  to (rounds - 1) do
6:   for  $j = 1$  to  $N$  in parallel do
7:      $\mathbf{X}_t^{(j)} \leftarrow \mathbf{X}_t$  ▷ Retrieve global model
8:     for  $k = 1$  to n_steps do
9:       Sample a batch  $\xi_{t,k}^{(j)}$ 
10:       $\nabla f_k \leftarrow \nabla f(\mathbf{X}_t^{(j)}; \xi_{t,k}^{(j)})$ 
11:       $\mathbf{G}_t^{(j)} \leftarrow \mathbf{G}_t^{(j)} + \nabla f_k$ 
12:    end for
13:
14:     $\mathbf{G}_t^{(j)} \leftarrow \mathbf{G}_t^{(j)} / \text{n\_steps}$  ▷ Gradient averaging
15:     $\mathbf{M}_t^{(j)} \leftarrow \mu \mathbf{M}_{t-1}^{(j)} + \mathbf{G}_t^{(j)}$  ▷ Local momentum update
16:     $\mathbf{Y}_t^{(j)} \leftarrow \text{ReshapeTo2D}(\mathbf{M}_t^{(j)})$  ▷ Reshape momentum to matrix  $m \times n$ 
17:    for  $k = 1$  to n_steps do ▷ Orthogonalization (Muon LMO)
18:       $\mathbf{A}_t^{(j)} \leftarrow \mathbf{Y}_t^{(j)} (\mathbf{Y}_t^{(j)})^\top$ 
19:       $\mathbf{Y}_t^{(j)} \leftarrow a \mathbf{Y}_t^{(j)} - b \mathbf{A}_t^{(j)} \mathbf{Y}_t^{(j)} + c (\mathbf{A}_t^{(j)})^2 \mathbf{Y}_t^{(j)}$ 
20:    end for
21:     $\mathbf{D}_t^{(j)} \leftarrow \text{ReshapeToOriginal}(\mathbf{Y}_t^{(j)})$ 
22:     $\mathbf{s}_t^{(j)} \leftarrow \text{sign}(\mathbf{D}_t^{(j)})$  ▷ Sign compression
23:    Send  $\mathbf{s}_t^{(j)}$  to server
24:  end for
25:   $\mathbf{s}_t^{\text{agg}} \leftarrow \text{sign}\left(\sum_{j=1}^N \mathbf{s}_t^{(j)}\right)$  ▷ Majority-vote aggregation on the server
26:
27:   $\mathbf{X}_{t+1} \leftarrow \mathbf{X}_t - \eta_t \mathbf{s}_t^{\text{agg}}$  ▷ Matrix parameter update
28:
29:   $\mathbf{X}_{t+1}^{\text{last}} \leftarrow \text{AdamW}(\mathbf{X}_t^{\text{last}}, \mathbf{s}_t^{\text{agg}}, \eta_t^{\text{aux}})$  ▷ Last layer update
30: end for
31: return  $\mathbf{X}_{\text{rounds}}$ 

```

A.3 Federated Algorithm with Error Feedback

Algorithm 3 Federated EF Sign-Muon

Input: Initial model $\mathbf{X}_0$, clients number N , rounds number rounds, local steps number n_steps, learning rate for the main layers η_t , learning rate for the last layer η_t^{aux} , momentum coefficient μ

Output: Updated global model $\mathbf{X}$

```

1:  $\mathbf{G}_0 \leftarrow 0$  ▷ Initialize the global gradient estimate on the server (for the main layers)
2: for  $j = 1$  to  $N$  do
3:    $\mathbf{g}_0^{(j)} \leftarrow 0, \mathbf{M}_0^{(j)} \leftarrow 0$  ▷ Initialize local estimators and momentum
4: end for
5: Initialize AdamW for last_layer with learning rate  $\eta_0^{\text{aux}}$ 
6: for  $t = 0$  to (rounds - 1) do
7:   for  $j = 1$  to  $N$  in parallel do ▷ Local step of EF21
8:      $\mathbf{X}_t^{(j)} \leftarrow \mathbf{X}_t$  ▷ Retrieve global model
9:      $\mathbf{G}_t^{(j)} \leftarrow 0$ 
10:    for  $k = 1$  to n_steps do
11:      Sample a batch  $\xi_{t,k}^{(j)}$ 
12:       $\nabla f_k \leftarrow \nabla f(\mathbf{X}_t^{(j)}; \xi_{t,k}^{(j)})$ 
13:       $\mathbf{G}_t^{(j)} \leftarrow \mathbf{G}_t^{(j)} + \nabla f_k$ 
14:    end for
15:     $\mathbf{G}_t^{(j)} \leftarrow \mathbf{G}_t^{(j)} / \text{n\_steps}$  ▷ Gradient averaging
16:     $\mathbf{M}_t^{(j)} \leftarrow \mu \mathbf{M}_{t-1}^{(j)} + \mathbf{G}_t^{(j)}$  ▷ Local momentum update
17:    for each parameters  $\mathbf{W}$  not in last_layer do
18:       $\Delta_t^{(j)} \leftarrow \mathbf{M}_t^{(j)} - \mathbf{g}_t^{(j)}$  ▷ Difference computation
19:       $\alpha_t^{(j)} \leftarrow \text{mean}(|\Delta_t^{(j)}|)$  ▷ Adaptive scaling factor
20:       $\mathbf{s}_t^{(j)} \leftarrow \text{sign}(\Delta_t^{(j)})$  ▷ Compress shifted gradient
21:       $\mathbf{g}_{t+1}^{(j)} \leftarrow \mathbf{g}_t^{(j)} + \alpha_t^{(j)} \mathbf{s}_t^{(j)}$  ▷ Compute gradient estimator
22:      Send  $(\mathbf{s}_t^{(j)}, \alpha_t^{(j)})$  to the server
23:    end for
24:    for each parameters  $\mathbf{W}^{\text{last}}$  in last_layer do
25:      Send  $\mathbf{G}_t^{(j)}$  to the server
26:    end for
27:  end for
28:   $\mathbf{G}_{t+1} \leftarrow \mathbf{G}_t + \frac{1}{N} \sum_{j=1}^N \alpha_t^{(j)} \mathbf{s}_t^{(j)}$  ▷ Update the global estimator on the server
29:  for each parameters  $\mathbf{X}$  do
30:     $\mathbf{Y} \leftarrow \text{ReshapeTo2D}(\mathbf{G}_{t+1})$ 
31:    for  $k = 1$  to n_steps do
32:       $\mathbf{A} \leftarrow \mathbf{Y} \mathbf{Y}^\top$ 
33:       $\mathbf{Y} \leftarrow a \mathbf{Y} - b \mathbf{A} \mathbf{Y} + c \mathbf{A}^2 \mathbf{Y}$ 
34:    end for
35:     $\mathbf{D}_{t+1} \leftarrow \text{ReshapeToOriginal}(\mathbf{Y})$ 
36:     $\mathbf{X}_{t+1} \leftarrow \mathbf{X}_t - \eta_t \mathbf{D}_{t+1}$  ▷ Update the matrix parameters
37:  end for
38:   $\nabla^{\text{last}} \leftarrow \frac{1}{N} \sum_{j=1}^N \mathbf{G}_t^{(j)}$ 
39:   $\mathbf{X}_{t+1}^{\text{last}} \leftarrow \text{AdamW}(\mathbf{X}_t^{\text{last}}, \nabla^{\text{last}}, \eta_t^{\text{aux}})$  ▷ Classifier update
40: end for
41: return  $\mathbf{X}_{\text{rounds}}$ 

```

A.4 Гипотезы исследования

Сформулируем две основные гипотезы для теоретического и экспериментального исследования предложенных алгоритмов. Пусть $\{\mathbf{X}_t\}_{t \geq 0}$ – последовательность состояний глобальной модели, а $f(\mathbf{X})$ – целевая функция в постановке федеративного обучения (уравнение 2).

Для анализа алгоритмов в обоих сценариях мы сделаем несколько предположений (Assumption).

Assumption 1 (Lower Boundedness). Целевая функция $f(\mathbf{X})$ ограничена снизу некоторым значением f^* , то есть $f(\mathbf{X}) \geq f^*$ для всех $\mathbf{X} \in \mathcal{X}$.

Assumption 2 (Smoothness). Функция $f(\mathbf{X})$ является L -гладкой относительно спектральной нормы $\|\cdot\|_2$. Это означает, что для любых $\mathbf{X}, \mathbf{Y} \in \mathcal{X}$ и дуальной ей нормы $\|\cdot\|_*$ выполняется неравенство:

$$\|\nabla f(\mathbf{X}) - \nabla f(\mathbf{Y})\|_* \leq L\|\mathbf{X} - \mathbf{Y}\|_2. \quad (14)$$

Assumption 3 (Stochastic Gradient). Стохастический градиент на каждом узле $j \in [N]$ является несмещенным и имеет ограниченную дисперсию в дуальной норме:

- *Unbiased:*

$$\mathbb{E}_{\xi \sim \mathcal{D}_j} [\nabla f_j(\mathbf{X}; \xi)] = \nabla f_j(\mathbf{X}), \quad (15)$$

- *Bounded variance:*

$$\mathbb{E}_{\xi \sim \mathcal{D}_j} [\|\nabla f_j(\mathbf{X}; \xi) - \nabla f_j(\mathbf{X})\|_*^2] \leq \sigma^2, \quad \sigma > 0. \quad (16)$$

Assumption 4 (Biased Compression). Введем ограничения на компрессор, следуя подходам из *Beznosikov et al. (2024)*. Существуют константы $\gamma > 0$ и $\beta > 0$ такие, что для любой матрицы $\mathbf{Y} \in \mathbb{R}^{m \times n}$ знаковый оператор $\mathcal{C}(\mathbf{Y}) = \text{sign}(\mathbf{Y})$ удовлетворяет условиям:

$$\gamma\|\mathbf{Y}\|_1 \leq \langle \mathbf{Y}, \text{sign}(\mathbf{Y}) \rangle = \|\mathbf{Y}\|_1, \quad (17)$$

$$\|\text{sign}(\mathbf{Y})\|_F^2 \leq \beta\|\mathbf{Y}\|_1. \quad (18)$$

Hypothesis 1 (Сходимость). При выполнении условий **Assumption 1-4** алгоритм *Federated SignMuon* сходится в гладком невыпуклом случае. Существует такой выбор learning rate η , при котором средняя норма градиента удовлетворяет оценке:

$$\min_{0 \leq t \leq T} \mathbb{E} [\|\nabla f(\mathbf{X}_t)\|_*^2] = \mathcal{O}(T^{-1/2}). \quad (19)$$

Это означает, что применение экстремальной 1-битной знаковой компрессии после матричной ортогонализации (LMO) не ухудшает асимптотическую скорость сходимости по сравнению с полноточными неевклидовыми методами в гладком невыпуклом случае.

Hypothesis 2 (Сохранение преимуществ Muon). Переход от LMO-направления Muon $\mathbf{D}_t = \mathbf{U}_t \mathbf{V}_t^\top$ к его знаковой компрессии $\text{sign}(\mathbf{D}_t)$ сохраняет основную геометрию ортонормированных обновлений. В результате *SignMuon* должен сохранять качество обучения, близкое к полному Muon, при существенно меньших коммуникационных затратах и превосходить *SignSGD* как в централизованном, так и в федеративном режимах.

A.5 Proof of Theorem 1

Proof The proof is constructed using a counterexample that shows the existence of a smooth function and a matrix dimension $N \geq 4$ for which the *SignMuon* algorithm is guaranteed to diverge. Extensive numerical searches strongly suggest that no counterexamples exist for $N \leq 3$ and *SignMuon* should converge. However, at $N = 4$, this convergence guarantee vanishes.

Consider a simple linear objective function $f(\mathbf{W}) = \text{Tr}(\mathbf{G}^\top \mathbf{W})$. For standard gradient descent or Muon, such a function diverges towards $-\infty$. The gradient is globally constant: $\nabla f(\mathbf{W}_t) = \mathbf{G}$ for all t . The change in the objective function at each iteration of the *SignMuon* algorithm is given by:

$$f(\mathbf{W}_{t+1}) - f(\mathbf{W}_t) = -\eta \langle \mathbf{G}, \text{sign}(\mathbf{U} \mathbf{V}^\top) \rangle. \quad (20)$$

To prove divergence, it is sufficient to present a matrix $\mathbf{G}$ for which $\langle \mathbf{G}, \text{sign}(\mathbf{U} \mathbf{V}^\top) \rangle < 0$. In this case, $f(\mathbf{W})$ strictly increases at every step, and the algorithm perpetually moves in the wrong direction.

We construct $\mathbf{G} \in \mathbb{R}^{4 \times 4}$ by defining a specific orthogonal matrix $\mathbf{O}$ and a specific rank-1 principal component $\mathbf{u}_1 \mathbf{v}_1^\top$. Let the orthogonal matrix $\mathbf{O}$ be given by the following exact rational numbers:

$$\mathbf{O} = \frac{1}{103} \begin{pmatrix} 101 & 20 & 2 & -2 \\ -20 & 97 & 20 & -20 \\ -2 & 20 & 2 & 101 \\ -2 & 20 & -101 & -2 \end{pmatrix}. \quad (21)$$

Because no entry is zero, its element-wise sign matrix $\mathbf{S} = \text{sign}(\mathbf{O})$ is uniquely defined. Now, let $\mathbf{u}_1$ and $\mathbf{v}_1$ be the following exact unit vectors:

$$\mathbf{u}_1 = \frac{1}{\sqrt{309}} \begin{pmatrix} 10 \\ -3 \\ 10 \\ 10 \end{pmatrix}, \quad \mathbf{v}_1 = \frac{1}{\sqrt{309}} \begin{pmatrix} 10 \\ 3 \\ -10 \\ 10 \end{pmatrix}. \quad (22)$$

One can easily verify that $\mathbf{O} \mathbf{v}_1 = \mathbf{u}_1$, meaning $\mathbf{u}_1$ and $\mathbf{v}_1$ act perfectly as left and right singular vectors for this orthogonal space. The crucial feature of this geometry is that the Frobenius inner product between this rank-1 component and the sign matrix $\mathbf{S}$ yields an exact, strictly negative fraction:

$$\langle \mathbf{u}_1 \mathbf{v}_1^\top, \mathbf{S} \rangle = \mathbf{u}_1^\top \mathbf{S} \mathbf{v}_1 = -\frac{43}{103}. \quad (23)$$

We construct the gradient matrix $\mathbf{G}$ by assigning a massive singular value ($\sigma_1 = 1000$) to this pathological component and a singular value of 1 to the rest of the orthogonal space:

$$\mathbf{G} = \begin{pmatrix} 324.6 & 97.3 & -323.6 & 323.6 \\ -97.3 & -28.2 & 97.3 & -97.3 \\ 323.6 & 97.3 & -323.6 & 324.6 \\ 323.6 & 97.3 & -324.6 & 323.6 \end{pmatrix}. \quad (24)$$

By definition of the Singular Value Decomposition (SVD), the leading singular vectors of $\mathbf{G}$ are $\mathbf{u}_1$ and $\mathbf{v}_1$ (with $\sigma_1 = 1001$), and the remaining singular values are $\sigma_2 = \sigma_3 = \sigma_4 = 1$. Consequently, the Muon orthogonalized step is exactly identical to $\mathbf{O}$, i.e., $\mathbf{U}_G \mathbf{V}_G^\top = \mathbf{O}$.

Substituting the resulting matrix $\mathbf{G}$ into the descent condition yields:

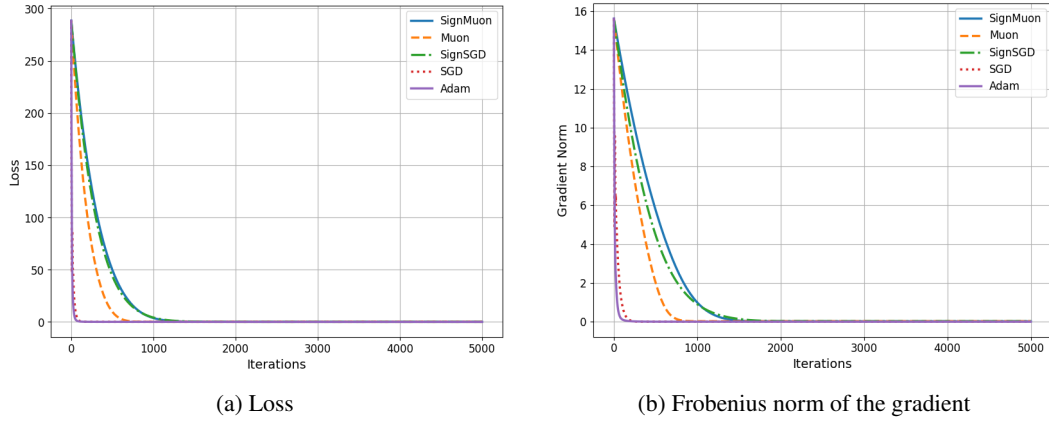
$$\langle \mathbf{G}, \mathbf{S} \rangle = \langle 1000 \mathbf{u}_1 \mathbf{v}_1^\top + \mathbf{O}, \mathbf{S} \rangle = 1000 \langle \mathbf{u}_1 \mathbf{v}_1^\top, \mathbf{S} \rangle + \langle \mathbf{O}, \mathbf{S} \rangle. \quad (25)$$

The inner product $\langle \mathbf{O}, \mathbf{S} \rangle$ is equivalent to the L_1 norm (sum of absolute values) of $\mathbf{O}$, which equals exactly $\frac{532}{103}$. Therefore:

$$\langle \mathbf{G}, \mathbf{S} \rangle = 1000 \left(-\frac{43}{103} \right) + \frac{532}{103} = -\frac{42468}{103} \approx -412.31. \quad (26)$$

Because $\langle \mathbf{G}, \mathbf{S} \rangle < 0$, the change in the objective function at every iteration is given by $f(\mathbf{W}_{t+1}) = f(\mathbf{W}_t) + 412.31\eta$. Thus, instead of minimizing, the algorithm strictly and monotonically diverges to $+\infty$, which completes the proof. ■

A.6 Results for the smooth convex problem

Рис. 2: Smooth convex problem (matrix formulation 12 for 500×500).

A.7 Centralized training results

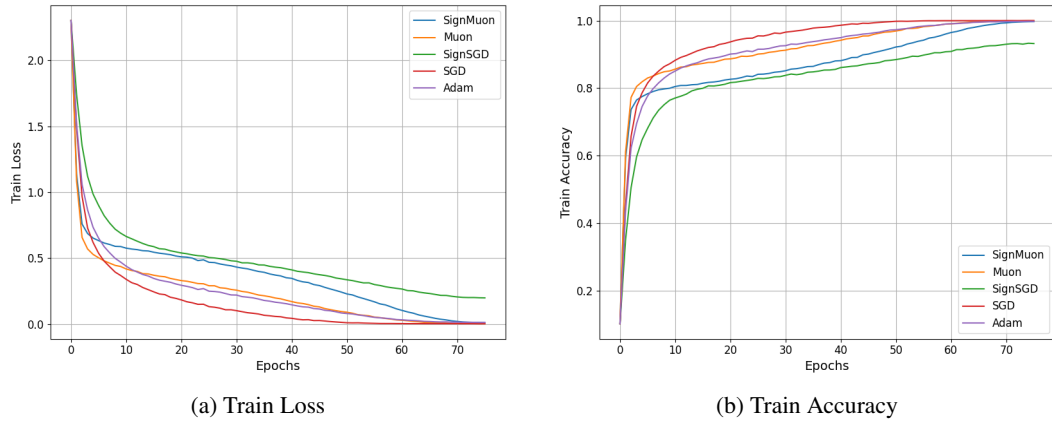


Рис. 3: Centralized setting. Comparison of convergence and classification accuracy achieved by different optimization algorithms during CIFAR-10 classifier training on the training set.

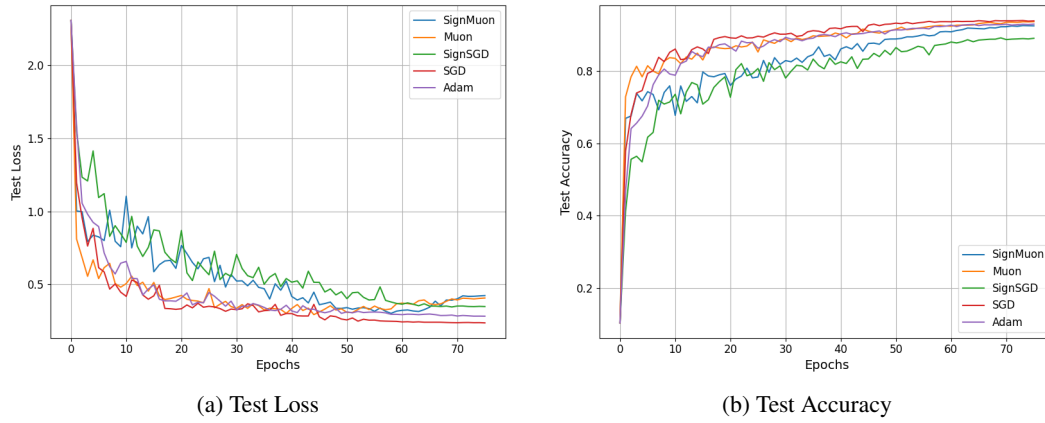


Рис. 4: Centralized setting. Comparison of convergence and classification accuracy achieved by different optimization algorithms during CIFAR-10 classifier training on the test set.

A.8 Federated training results. Experiment 1

Таблица 5: CIFAR-10 Federated Learning. Experiment 1. Scalability of the SignMuon Algorithm.

Dataset	Optimizer	Rounds	Clients	Steps	Batch	LR	Mom	Test Acc
CIFAR-10	SignMuon	2000	1	1	64	0.001	0.9	76.13%
			3					81.24%
			5					82.87%
			7					83.54%
			9					84.03%

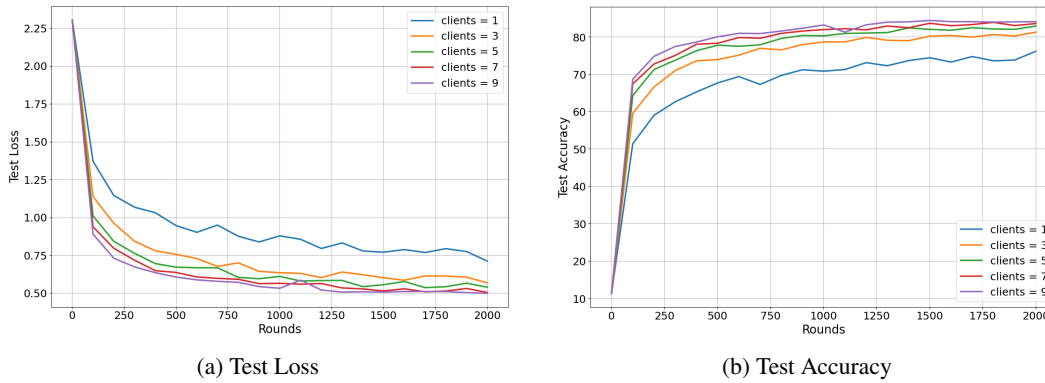
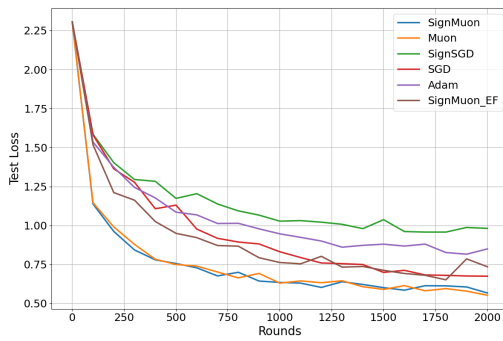


Рис. 5: CIFAR-10 Federated Learning. Experiment 1. Scalability of the SignMuon Algorithm.

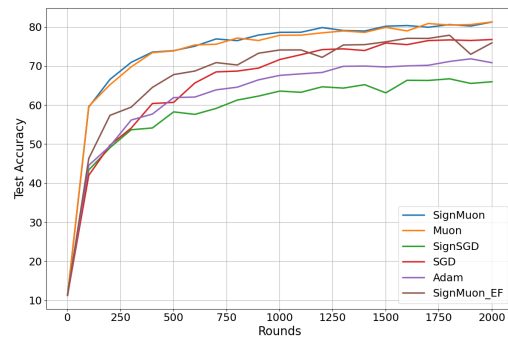
A.9 Federated training results. Experiment 2

Таблица 6: CIFAR-10 Federated Learning. Experiment 2: Comparison of Optimization Algorithms with (N = 3) Clients.

Dataset	Optimizer	Rounds	Clients	Steps	Batch	LR	Mom	Test Acc
CIFAR-10	SignMuon	2000	3	1	64	0.001	0.9	81.24%
	Muon					0.05		81.27%
	SignSGD					0.001		65.96%
	SGD					0.03		76.77%
	Adam					0.001		70.83%
	SignMuon_EF					0.02		75.92%



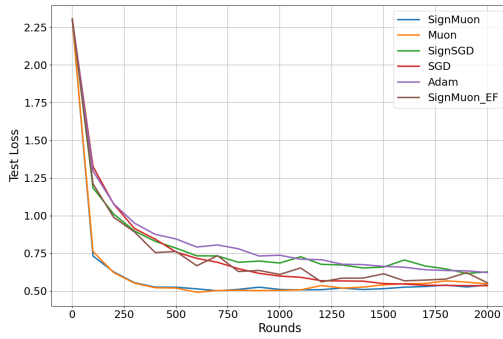
(a) Test Loss



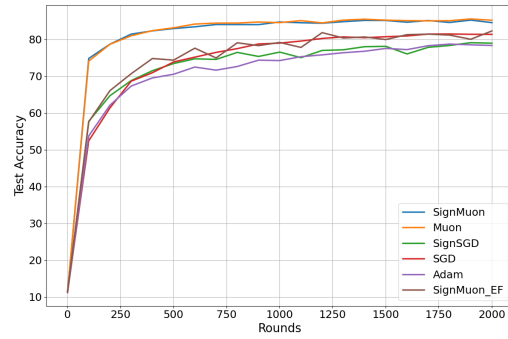
(b) Test Accuracy

Рис. 6: CIFAR-10 Federated Learning. Experiment 2: Comparison of Optimization Algorithms with (N = 3) Clients.

A.10 Federated training results. Experiment 3



(a) Test Loss



(b) Test Accuracy

Рис. 7: CIFAR-10 Federated Learning. Experiment 3: Comparison of Optimization Algorithms with (N = 10) Clients.